

Operator Discovery Report: bounded_worse_threshold_acceptance_stagnation_bias

- Operator type: acceptance.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministically allows a small, structure-agnostic amount of bounded-worse 2-opt/edge-relink moves to break 2-opt plateaus; acceptance tightens with late-search cooling and relaxes slightly under short stagnation burst
- Novelty classification: threshold_acceptance.
- Rediscovery signature: acceptance:threshold:thr=0.018:temp=0.005.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

Pseudocode

- Use threshold acceptance when a restart solution is not immediately improving.
- Relax or tighten acceptance depending on stagnation and late-search cooling.
- Reject candidate restarts that exceed the configured worse-move budget.

Complexity

- Non-empty lines: 11.
- Complexity score: 0.42.
- Fingerprint: adef96fd584bc090.

Why It Should Help

- The operator only changes restart acceptance, so any gain is attributable to a narrow exploration-vs-stability decision rule.

Failure Cases

- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 111.858443, distance-eval delta 0.0.
- cheapest_insertion_2opt gap delta 0.0, runtime delta 502.796929, distance-eval delta 0.0.
- random_restart_2opt gap delta 0.0, runtime delta 2.1704, distance-eval delta 252.0.
- sparse_three_opt gap delta 0.0, runtime delta 6.232029, distance-eval delta 0.0.
- clustered_local_search gap delta -0.000666, runtime delta -9.484543, distance-eval delta 1594.285714.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.
- clustered_tsp operator gap 0.110853 vs baseline 0.110853.

- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: 0.0.
- Runtime inflation: 0.013898.
- Distance-evaluation inflation: 0.0.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.0.
- improving_only_acceptance mean gap 0.121317 and delta vs full operator 0.0.
- tight_threshold_acceptance mean gap 0.121317 and delta vs full operator 0.0.